

ADITHYA BHAT

Visa Research, Visa Inc., CA

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◇ Website - <https://adityabhatkajake.github.io>

RESEARCH INTERESTS

My interests are *Byzantine fault-tolerant distributed systems* and *applied cryptography*. My work aims at developing and evaluating cryptographic solutions for secure, fault-tolerant distributed systems. My current research focuses on energy-efficient consensus protocols in network settings such as synchronous, asynchronous, and all intermediate models; fault-tolerant cryptographic protocols such as PVSS and Random Beacons and Private set intersection.

EDUCATION

Purdue University, West Lafayette

(2018-2023)

Advisor: Aniket Kate

Ph.D., Department of Computer Science

National Institute of Technology Karnataka, Surathkal

(2014-2018)

Bachelor of Technology, Information Technology

WORK EXPERIENCE

Visa Research, Foster City

(Aug 2023 - Present)

Staff Research Scientist

(Anderson Nascimento)

Visa Research, Palo Alto

(May 2022 - Aug 2022)

Ph.D. Research Intern

(Mahdi Zamani)

- Developed *FastSync*: an efficient blockchain synchronization protocol
- Developed an efficient partially synchronous sharding protocol and implemented Instachain

VMware Research, Remote

(May 2021 - Aug 2021)

Research Intern

(Alin Tomescu, Ittai Abraham)

- Built a prototype of anonymous token system using Concord-BFT
- Developed *quick-pay*: a one-round trip low-latency payment system
- Developed a two-phase lock-free sharding solution using quick-pay

Purdue University, West Lafayette

(Aug 2018 - Aug 2023)

Graduate Research Assistant

(Aniket Kate)

- Researching energy efficient Byzantine fault tolerant consensus protocols.
- Developed mathematical models for protocol optimization to improve energy efficiency.
- Implemented, evaluated, and simulated cryptographic and distributed system protocols.

Indian Statistical Institute, Kolkata

(July 2017 - December 2017)

Undergraduate Research Fellow

(Sushmitha Ruja)

- Designed a storage auditing library based on compact proofs of retrievability.
- Implemented new transactions on an Ethereum client to build a publicly verifiable data-storage system.

Morgan Stanley, Bangalore

(May 2017 - July 2017)

Software Analyst

- Worked on evaluating an elastic-search visualization plugin for the LevelDB database.

Indian Institute of Science, Bangalore

(May 2016 - July 2016)

Indian Academy of Sciences Summer Research Fellow

(*C. E. Veni Madhavan*)

- Evaluated Pollard's rho, William's $p + 1$, $p - 1$ factorization, and elliptic curve factorization methods for efficient batch factorization of numbers generated during the sieving phase of GGNFS.

PUBLICATIONS

1. *Lite-PoT: Practical Powers-of-Tau Setup Ceremony*. **CCS 2025**. Lucien K. L. Ng, Pedro Moreno-Sanchez, Mohsen Minaei, Panagiotis Chatzigiannis, **Adithya Bhat**, Duc V. Le. [eprint](#)
2. *Synchronous Distributed Key Generation without Broadcasts*. **Journal of Cryptology 2024**. Nibesh Shrestha, **Adithya Bhat**, Kartik Nayak, Aniket Kate. [journal eprint](#)
3. *SensorBFT: Fault-Tolerant Target Localization using Voronoi Diagrams and Approximate Agreement*. **ICDCS 2024**. Akhil Bandrupalli, **Adithya Bhat**, Somali Chatterji, Michael K. Reiter, Aniket Kate, Saurabh Bagchi. [conference eprint](#)
4. *Delphi: Efficient Asynchronous Approximate Agreement for Distributed Oracles*. **DSN 2024**. Akhil Bandrupalli, **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, Chen-Da Liu-Zhang, and Michael K. Reiter. [conference eprint](#)
5. *HashRand: Efficient Asynchronous Random Beacon without Threshold Cryptographic Setup*. **CCS 2024**. Akhil Bandrupalli, **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, and Michael Reiter. [conference eprint](#)
6. *Attacking and Improving the Tor Directory Protocol*. **IEEE S&P 2024**. Zhongtang Luo, **Adithya Bhat**, Kartik Nayak, and Aniket Kate. [conference eprint](#)
7. *EESMR - Energy Efficient State Machine Replication*. **Middleware 2023**. **Adithya Bhat**, Akhil Bandrupalli, Manish Nagaraj, Saurabh Bagchi, Aniket Kate, Michael Reiter. [conference eprint](#)
8. *The unique chain rule and its applications*. **Financial Cryptography 2023**. **Adithya Bhat**, Akhil Bandrupalli, Saurabh Bagchi, Aniket Kate, Michael Reiter. [pre-conference](#) [conference eprint](#) [code](#)
9. *OptRand - Optimistically Responsive Reconfigurable Distributed Randomness*. **NDSS 2023**. **Adithya Bhat**, Nibesh Shrestha, Aniket Kate, Kartik Nayak. [conference eprint](#) [protocol](#) [code](#) [crypto](#) [code](#) [video](#)
10. *OpenSquare: Decentralized Repeated Modular Squaring Service*. **CCS 2021**. Sri Aravinda Krishnan Thyagarajan, Tiantian Gong, **Adithya Bhat**, Aniket Kate, Dominique Schroder. [conference eprint](#) [code](#)
11. *RandPiper - Reconfiguration Friendly Random Beacons with Quadratic Communication*. **CCS 2021**. **Adithya Bhat**, Nibesh Shrestha, Aniket Kate, Kartik Nayak. [conference eprint](#) [code](#)
12. *Reparo - Publicly Verifiable Repair Layer for any Blockchain*. **FC 2021**. Sri Aravinda Krishnan Thyagarajan, **Adithya Bhat**, Bernardo Magri, Daniel Tschudi, Aniket Kate. [conference eprint](#)
13. *Verifiable Timed Signatures for Blockchains*. **CCS 2020**. Sri Aravinda Krishnan Thyagarajan, **Adithya Bhat**, Giulio Malavolta, Nico Döttling, Aniket Kate, Dominique Schroder. [conference eprint](#) [code](#)

PATENTS

1. *Accountable Decentralized Anonymous Payments*. United States Patent Application 20240265373. [patent](#).
2. *Fast sync blockchain system and method*. United States Patent Application WO2024026321A1. [patent](#).
3. *A system for partially synchronous scalable blockchains*. United States Patent Application WO2024233453A1. [patent](#)

TECH REPORTS

1. Ira: Efficient Transaction Replay for Distributed Systems. **Adithya Bhat**, Harshal B Shah, Mohsen Minaei. [eprint](#)
2. FPS: Flexible Payment Systems. **Adithya Bhat**, Srinivasan Raghuraman, Panagiotis Chatzigiannis, Duc V Le, Mohsen Minaei. [eprint](#)
3. SoK: *Fully-homomorphic encryption in smart contracts*. **Poster - FHE.org workshop**. Daniel Aronoff, **Adithya Bhat**, Panagiotis Chatzigiannis, Mohsen Minaei, Srinivasan Raghuraman, Robert M. Townsend, Nicolas Xuan-Yi Zhang. [eprint](#)
4. *Leto - Partially Synchronous Unique Chains made flexible*. **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, Michael Reiter. [code](#)
5. *UTT: Decentralized Ecash with Accountable Privacy*. **Science of Blockchain Conference 2023**. Alin Tomescu, Adithya Bhat, Benny Applebaum, Ittai Abraham, Guy Gueta, Benny Pinkas, and Avishay Yanai. [eprint](#) [code](#)
6. *Transitive Network - A Tokenless IOU-based Credit Network*. **Cryptocurrency Implementers Workshop, FC 2019**. **Adithya Bhat**, Pedro Moreno Sanchez, Aniket Kate. [code](#)

SOFTWARE ARTIFACTS

1. Developed a synchronous networking library to implement SMR protocols. [code](#) (Rust)
2. Implemented Apollo [8] (protocol node, normal client and special client) using the Rust networking library. [code](#) (Rust)
3. Implemented Sync HotStuff (normal protocol node, round robin protocol node, client) using the Rust networking library. [code](#) (Rust)
4. Developed a plug-and-play framework using libp2p to run and simulate distributed system protocols. The framework provides interfaces to aid faster prototyping of distributed system protocols. [code](#) (Go-lang)
5. Implemented Sync HotStuff using the go networking library. [code](#) (Go-lang)
6. Implemented Apollo [8] using the go networking library. [code](#) (Go-lang)
7. Implementation of E2C [7]. [code](#) (C++)
8. Developed a linearly homomorphic time-lock puzzle library. [code](#) (C)

TALKS

1. Attacking and Improving the Tor Directory Protocol. (RWC 2025)

2. A study of practical BFT systems. *(Supra Team)*
3. A study of practical BFT systems. *(Espresso Systems)*
4. A study of practical BFT systems. [abstract](#) *(Next Generation Secure Distributed Computing Seminar, Dagstuhl)*
5. OptRand, an optimistically responsive distributed random beacon protocol. [video](#) *(Web3 Randomness Workshop)*
6. Reconfiguration-friendly Byzantine Fault-tolerant Distributed randomness. [slides](#) *(KU Leuven)*
7. Unique Chain Rule and its applications. [slides](#) *(FC 2023)*
8. Reconfiguration-friendly Byzantine Fault-tolerant Distributed randomness. [slides](#) *(Boston University)*
9. Flexible State Machine Replication. [slides](#) *(Midwest Crypto Day - Lightning session)*
10. OptRand - Optimistically Responsive Reconfigurable Distributed Randomness. [video](#) *(NDSS 2023)*
11. FastSync: Using the future to verify the past. [video](#) *(CESC 2022)*
12. RandPiper - Reconfiguration friendly random beacons with quadratic communication. *(CCS 2021)*
13. Reparo - Publicly Verifiable Repair Layer for any blockchain. [video](#) *(FC 2021)*
14. Transitive network - A tokenless IOU-based Credit Network. Cryptocurrency Implementers Workshop. *(FC 2019)*

ACADEMIC SERVICE

- Program Committee:
 - 2026: FC, CCS, ACNS, Usenix Security
 - 2025: FC, CCS, CVC
 - 2024: CCS
- Reviewer for
 - 2024: ACM TOPS, Journal of Cryptology
 - 2023: SOSP AEC, IET
- External Reviewer for
 - 2025: ITCS, Crypto
 - 2024: IEEE S & P
 - 2023: CCS, IET, Middleware, IEEE S & P
 - 2022: CCS, PODC, IEEE S & P, CESC
 - 2021: AFT, FC, PODC, IEEE S & P
 - 2020: Usenix Security, IEEE S & P
 - 2019: NDSS